

PES University - Dept. of CSE

Lab 1: Exception Use-Case Flow Specification

PROBLEM STATEMENT #43: Feature Flag & Dynamic Config Manager

Use Case ID:	UC-01 (Exceptions)
Use Case Name:	Evaluate Feature Flag & Dynamic Config (Exception Flows)
Primary Actor:	Client SDK / Application (also Software Engineer)
Preconditions:	1. Client SDK is initialized. 2. Evaluation requests are sent by Client SDK, but preconditions for normal processing fail (e.g. invalid API Key, unauthorized environment access, or database offline with empty cache).
Postconditions:	1. System terminates flow and returns appropriate HTTP error response (401/403/500). 2. Security, validation, or system failure telemetry events are recorded.

Exception Flow Specifications

1. Exception Flow 1b (HTTP 401): Invalid or Expired API Key

- 1b1.** Client SDK sends an evaluation request specifying flag_key, user_id, and user_attributes but provides an expired or invalid API key.
- 1b2.** System attempts to validate the client's API key context within the authorization database and fails validation.
- 1b3.** System logs an authentication failure security event, recording the client IP, invalid API key prefix, and timestamp.
- 1b4.** System returns HTTP 401 Unauthorized with an error payload: { "error": "Unauthorized", "message": "Invalid or Expired API Key" }.
- 1b5.** Client SDK catches the HTTP 401 response, halts evaluation, throws a fatal initialization exception to the host application, and the use case terminates unsuccessfully.

2. Exception Flow 1c (HTTP 403): Environment Access Denied

- 1c1.** Client SDK sends an evaluation request targeting a specific environment (e.g., Production).
- 1c2.** System validates the client's API key but determines that the API key's scope restricts access only to Development or Staging, and not Production.
- 1c3.** System logs an environment access authorization violation security alert in the audit database.
- 1c4.** System returns HTTP 403 Forbidden with an error payload: { "error": "Forbidden", "message": "Access Denied to Environment" }.
- 1c5.** Client SDK catches the HTTP 403 response, logs a permissions failure telemetry event, falls back to the hardcoded developer fallback value, and the use case terminates.

3. Exception Flow 1d (HTTP 500 & Empty Cache): Database Outage & Empty Cache

- 1d1.** Client SDK sends an evaluation request payload specifying flag_key, user_id, and user_attributes.
- 1d2.** System attempts to retrieve active feature flag schemas and targeting rules from the main database, but database connections fail (network timeout or database service offline), resulting in an internal server error.
- 1d3.** System attempts to retrieve rules from backup edge Redis memory cache but fails, returning HTTP 500 Internal Server Error to the Client SDK.
- 1d4.** Client SDK detects the HTTP 500 error and queries its persistent local disk/memory storage for the last valid rules cache.
- 1d5.** Client SDK determines that the local rules cache is empty (e.g., first-time application boot or cache expired).
- 1d6.** SDK applies the hardcoded developer fallback value, logs a critical system failure warning to local diagnostics, schedules a background synchronization retry task, and the use case completes successfully with the fallback state.